ELEC 2FH4 - FINAL CHEAT SHEET

Amaiam Ul Haque

TABLE 2.1 Relationships between Rectangular, Cylindrical, and Spherical Coordinates

Rectangular to Cylindrical

Variable change $\begin{cases} x = \rho\cos\phi \\ y = \rho\sin\phi \\ z = z \end{cases}$

Component change $\begin{cases} A_\rho = A_x\cos\phi + A_y\sin\phi \\ A_\phi = -A_x\sin\phi + A_y\cos\phi \\ A_z = A_z \end{cases}$

Cylindrical to Rectangular

Variable change $\begin{cases} \rho = \sqrt{x^2+y^2} \\ \phi = \tan^{-1}\left(\frac{y}{x}\right) \\ z = z \end{cases}$ $\begin{cases} \sin\phi = \frac{y}{\sqrt{x^2+y^2}} \\ \cos\phi = \frac{x}{\sqrt{x^2+y^2}} \end{cases}$

Component change $\begin{cases} A_x = A_\rho\frac{x}{\sqrt{x^2+y^2}} - A_\phi\frac{y}{\sqrt{x^2+y^2}} \\ A_y = A_\rho\frac{y}{\sqrt{x^2+y^2}} + A_\phi\frac{x}{\sqrt{x^2+y^2}} \\ A_z = A_z \end{cases}$

Rectangular to Spherical

Variable change $\begin{cases} x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \end{cases}$

Component change $\begin{cases} A_r = A_x\sin\theta\cos\phi + A_y\sin\theta\sin\phi + A_z\cos\theta \\ A_\theta = A_x\cos\theta\cos\phi + A_y\cos\theta\sin\phi - A_z\sin\theta \\ A_\phi = -A_x\sin\phi + A_y\cos\phi \end{cases}$

Spherical to Rectangular

Variable change $\begin{cases} r = \sqrt{x^2+y^2+z^2} \\ \theta = \cos^{-1}\frac{z}{\sqrt{x^2+y^2+z^2}} \\ \phi = \tan^{-1}\left(\frac{y}{x}\right) \end{cases}$ $\begin{cases} \cos\theta = \frac{z}{\sqrt{x^2+y^2+z^2}} \\ \sin\theta = \frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}} \end{cases}$ $\begin{cases} \cos\phi = \frac{x}{\sqrt{x^2+y^2}} \\ \sin\phi = \frac{y}{\sqrt{x^2+y^2}} \end{cases}$

Component change $\begin{cases} A_x = \frac{A_r x}{\sqrt{x^2+y^2+z^2}} + \frac{A_\theta xz}{\sqrt{(x^2+y^2)(x^2+y^2+z^2)}} - \frac{A_\phi y}{\sqrt{x^2+y^2}} \\ A_y = \frac{A_r y}{\sqrt{x^2+y^2+z^2}} + \frac{A_\theta yz}{\sqrt{(x^2+y^2)(x^2+y^2+z^2)}} + \frac{A_\phi x}{\sqrt{x^2+y^2}} \\ A_z = \frac{A_r z}{\sqrt{x^2+y^2+z^2}} - \frac{A_\theta\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}} \end{cases}$

$$d\mathbf{l} = dx\,\mathbf{a}_x + dy\,\mathbf{a}_y + dz\,\mathbf{a}_z$$
$$d\mathbf{l} = d\rho\,\mathbf{a}_\rho + \rho\,d\phi\,\mathbf{a}_\phi + dz\,\mathbf{a}_z$$
$$d\mathbf{l} = dr\,\mathbf{a}_r + r\,d\theta\,\mathbf{a}_\theta + r\sin\theta\,d\phi\,\mathbf{a}_\phi$$

$$d\mathbf{S} = dy\,dz\,\mathbf{a}_x \quad dx\,dz\,\mathbf{a}_y \quad dx\,dy\,\mathbf{a}_z$$
$$d\mathbf{S} = \rho\,d\phi\,dz\,\mathbf{a}_\rho \quad d\rho\,dz\,\mathbf{a}_\phi \quad \rho\,d\rho\,d\phi\,\mathbf{a}_z$$
$$d\mathbf{S} = r^2\sin\theta\,d\theta\,d\phi\,\mathbf{a}_r \quad r\sin\theta\,dr\,d\phi\,\mathbf{a}_\theta \quad r\,dr\,d\theta\,\mathbf{a}_\phi$$

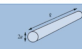
$$dv = dx\,dy\,dz$$
$$dv = \rho\,d\rho\,d\phi\,dz$$
$$dv = r^2\sin\theta\,dr\,d\theta\,d\phi$$

TABLE 8.1 Force on a Charged Particle

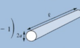
State of Particle	E Field	B Field	Combined E and B Fields
Stationary	$Q\mathbf{E}$	—	$Q\mathbf{E}$
Moving	$Q\mathbf{E}$	$Q\mathbf{u}\times\mathbf{B}$	$Q(\mathbf{E}+\mathbf{u}\times\mathbf{B})$

TABLE 8.3 A Collection of Formulas for Inductance of Common Elements

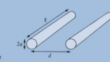
1. Wire
$L = \frac{\mu_o\ell}{8\pi}$



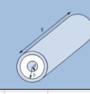
2. Hollow cylinder
$L = \frac{\mu_o\ell}{2\pi}\left(\ln\frac{2\ell}{a} - 1\right)$
$\ell \gg a$



3. Parallel wires
$L = \frac{\mu_o\ell}{\pi}\ln\frac{d}{a}$
$\ell \gg d,\ d \gg a$

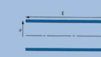


4. Coaxial conductor
$L = \frac{\mu_o\ell}{2\pi}\ln\frac{b}{a}$

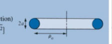


5. Circular loop
$L = \frac{\mu_o\ell}{2\pi}\left(\ln\frac{4\ell}{d} - 2.45\right)$
$\ell = 2\pi\rho_o,\ \rho_o \gg d$

6. Solenoid
$L = \frac{\mu_o N^2 S}{\ell}$
$\ell \gg a$



7. Torus (of circular cross section)
$L = \mu_o N^2[\rho_o - \sqrt{\rho_o^2 - a^2}]$



8. Sheet
$L = \mu_o 2\ell\left(\ln\frac{2\ell}{b+t} + 0.5\right)$

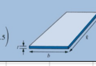


TABLE 8.4 Analogy between Electric and Magnetic Circuits

Electric	Magnetic
Conductivity σ	Permeability μ
Field intensity E	Field intensity H
Current $I = \int \mathbf{J}\cdot d\mathbf{S}$	Magnetic flux $\Psi = \int \mathbf{B}\cdot d\mathbf{S}$
Current density $J = \frac{I}{S} = \sigma E$	Flux density $B = \frac{\Psi}{S} = \mu H$
Electromotive force (emf) V	Magnetomotive force (mmf) $\mathcal{F}$
Resistance R	Reluctance $\mathcal{R}$
Conductance $G = \frac{1}{R}$	Permeance $\mathcal{P} = \frac{1}{\mathcal{R}}$
Ohm's law $R = \frac{V}{I} = \frac{\ell}{\sigma S}$ or $V = E\ell = IR$	Ohm's law $\mathcal{R} = \frac{\mathcal{F}}{\Psi} = \frac{\ell}{\mu S}$ or $\mathcal{F} = H\ell = \Psi\mathcal{R} = NI$
Kirchhoff's laws: $\sum I = 0$, $\sum V - \sum RI = 0$	Kirchhoff's laws: $\sum \Psi = 0$, $\sum \mathcal{F} - \sum \mathcal{R}\Psi = 0$

Chapter 1
- $A \cdot B$ - $AB\cos\theta_{AB}$
- $A \times B$ - $AB\sin\theta_{AB}\,\vec{a}_n$
- $A_B = A\cdot\vec{a}_B$
- $\vec{A}_B = A_B\,\vec{a}_B$

Chapter 2
- (x, y, z)
- (ρ, ϕ, z)
- (r, θ, ϕ)

Chapter 3
- gradient: ∇V
- divergence: $\nabla\cdot A$
- curl: $\nabla\times A$
- laplacian: $\nabla^2 V$

Theorems
- divergence: $\oint_S A\cdot ds = \int_v \nabla\cdot A\,dv$
- Stokes: $\oint_L A\cdot dl = \int_S (\nabla\times A)\,ds$
- laplaces: $0 = \nabla^2 V$

- Solenoidal field if $0 = \nabla\cdot A$
- irrotational field if $0 = \nabla\times A$

- $0 = \nabla\cdot\nabla\times A$
- $0 = \nabla\times\nabla V$

Chapter 4

$$\mathbf{F} = \frac{Q_1Q_2}{4\pi\varepsilon_o R^2}\mathbf{a}_R$$

$$\mathbf{E} = \frac{Q}{4\pi\varepsilon_o R^2}\mathbf{a}_R = \frac{Q\,\mathbf{R}}{4\pi\varepsilon_o R^3} \quad \text{(point charge only)}$$

$Q = \int_L \rho_L\,dl$ for line charge | $\mathbf{E} = \frac{\rho_L}{2\pi\varepsilon_o\rho}\mathbf{a}_\rho$, $\infty\ \rho_L$

$Q = \int_S \rho_S\,dS$ for surface charge | $\mathbf{E} = \frac{\rho_S}{2\varepsilon_o}\mathbf{a}_n$, $\infty\ \rho_S$

$Q = \int_v \rho_v\,dv$ for volume charge

$$\mathbf{D} = \varepsilon_o\mathbf{E}$$
$$\Psi = \int_S \mathbf{D}\cdot d\mathbf{S}$$
$$\Psi = \oint_S \mathbf{D}\cdot d\mathbf{S} = Q_{enc} = \int_v \rho_v\,dv$$

$\rho_v = \nabla\cdot\mathbf{D}$ (first Maxwell equation to be derived)

$$D_n\oint_S dS = Q_{enc} \quad \text{or} \quad D_n = \frac{Q_{enc}}{S}$$
$$W = -Q\int_A^B \mathbf{E}\cdot d\mathbf{l}$$
$$V(\mathbf{r}) = \frac{Q}{4\pi\varepsilon_o|\mathbf{r}-\mathbf{r}'|} + C$$
$$V = -\int_L \mathbf{E}\cdot d\mathbf{l} + C$$
$$V_{AB} = -\int_A^B \mathbf{E}\cdot d\mathbf{l} = \frac{W}{Q} = V_B - V_A$$
$$\oint_L \mathbf{E}\cdot d\mathbf{l} = 0$$

$\nabla\times\mathbf{E} = \mathbf{0}$ (second Maxwell equation to be de

$$V(\mathbf{r}) = \frac{\mathbf{p}\cdot(\mathbf{r}-\mathbf{r}')}{4\pi\varepsilon_o|\mathbf{r}-\mathbf{r}'|^3}$$
$$\mathbf{E} = -\nabla V$$
$$W_E = \frac{1}{2}\sum_{k=1}^n Q_k V_k$$
$$W_E = \frac{1}{2}\int_v \mathbf{D}\cdot\mathbf{E}\,dv = \frac{1}{2}\int_v \varepsilon_o|\mathbf{E}|^2\,dv$$

Chapter 5

$$I = \int \mathbf{J}\cdot d\mathbf{S}$$
$$R = \frac{\ell}{\sigma S}$$
$$\nabla\cdot\mathbf{J} + \frac{\partial\rho_v}{\partial t} = 0$$
$$E_{1t} = E_{2t}$$
$$D_{1n} - D_{2n} = \rho_S \quad \text{or} \quad D_{1n} = D_{2n} \quad \text{if} \quad \rho_S = 0$$
$$E_t = 0, \quad D_n = \varepsilon E_n = \rho_S$$

4. The macroscopic effect of polarization on a given volume of a dielectric material is to "paint" its surface with a bound charge $Q_b = \oint_S \rho_{ps}\,dS$ and leave within it an accumulation of bound charge $Q_b = \int_v \rho_{pv}\,dv$, where $\rho_{ps} = \mathbf{P}\cdot\mathbf{a}_n$ and $\rho_{pv} = -\nabla\cdot\mathbf{P}$.
5. In a dielectric medium, the **D** and **E** fields are related as $\mathbf{D} = \varepsilon\mathbf{E}$, where $\varepsilon = \varepsilon_o\varepsilon_r$ is the permittivity of the medium while **E** and **P** are related as $\mathbf{P} = \chi_e\varepsilon_o\mathbf{E}$.
6. The electric susceptibility $\chi_e (= \varepsilon_r - 1)$ of a dielectric measures the sensitivity of the material to an electric field.

Chapter 6

$$\boxed{\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0}$$

$$\boxed{\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial V}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial\phi^2} + \frac{\partial^2 V}{\partial z^2} = 0}$$

$$\boxed{\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial V}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 V}{\partial\phi^2} = 0}$$

$$\nabla\cdot\varepsilon\nabla V = -\rho_v$$
$$\nabla^2 V = -\frac{\rho_v}{\varepsilon}$$
$$\nabla^2 V = 0$$

5. The problem of finding the resistance R of an object or the capacitance C of a capacitor may be treated as a boundary-value problem. To determine R, we assume a potential difference V_o between the ends of the object, solve Laplace's equation, find $I = \int_S \sigma\mathbf{E}\cdot d\mathbf{S}$, and obtain $R = V_o/I$. Similarly, to determine C, we assume a potential difference of V_o between the plates of the capacitor, solve Laplace's equation, find $Q = \int_S \varepsilon\mathbf{E}\cdot d\mathbf{S}$, and obtain $C = Q/V_o$.

Chapter 7

TABLE 7.1 Analogy between Electric and Magnetic Fields*

Term	Electric	Magnetic
Basic laws	$\mathbf{F} = \frac{Q_1Q_2}{4\pi\varepsilon R^2}\mathbf{a}_R$	$d\mathbf{B} = \frac{\mu_o I\,d\mathbf{l}\times\mathbf{a}_R}{4\pi R^2}$
	$\oint \mathbf{D}\cdot d\mathbf{S} = Q_{enc}$	$\oint \mathbf{H}\cdot d\mathbf{l} = I_{enc}$
Force law	$\mathbf{F} = Q\mathbf{E}$	$\mathbf{F} = Q\mathbf{u}\times\mathbf{B}$
Source element	dQ	$dQ\mathbf{u} = I\,d\mathbf{l}$
Field intensity	$E = \frac{V}{\ell}$ (V/m)	$H = \frac{I}{\ell}$ (A/m)
Flux density	$D = \frac{\psi}{S}$ (C/m²)	$B = \frac{\psi}{S}$ (Wb/m²)
Relationship between fields	$\mathbf{D} = \varepsilon\mathbf{E}$	$\mathbf{B} = \mu\mathbf{H}$
Potentials	$\mathbf{E} = -\nabla V$	$\mathbf{H} = -\nabla V_m\ (\mathbf{J} = 0)$
	$V = \int_L \frac{\rho_L\,dl}{4\pi\varepsilon R}$	$\mathbf{A} = \int_L \frac{\mu I\,d\ell}{4\pi R}$
Flux	$\psi = \int \mathbf{D}\cdot d\mathbf{S}$	$\psi = \int_S \mathbf{B}\cdot d\mathbf{S}$
	$\psi = Q = CV$	$\psi = LI$
	$I = C\frac{dV}{dt}$	$V = L\frac{dI}{dt}$
Energy density	$w_E = \frac{1}{2}\mathbf{D}\cdot\mathbf{E}$	$w_m = \frac{1}{2}\mathbf{B}\cdot\mathbf{H}$
Poisson's equation	$\nabla^2 V = -\frac{\rho_v}{\varepsilon}$	$\nabla^2\mathbf{A} = -\mu\mathbf{J}$

$$d\mathbf{H} = \frac{I\,d\mathbf{l}\times\mathbf{R}}{4\pi R^3} \quad \text{(in A/m)}$$
$$I\,d\mathbf{l} \equiv \mathbf{K}\,dS \equiv \mathbf{J}\,dv$$
$$\oint_L \mathbf{H}\cdot d\mathbf{l} = I_{enc} = \int_S \mathbf{J}\cdot d\mathbf{S}$$

$\nabla\times\mathbf{H} = \mathbf{J}$ (third Maxwell equation to be derived)

current distribution is symmetric so that an Amperian path (on which $\mathbf{H} = H_\phi\mathbf{a}_\phi$ stant) can be found, Ampère's law is useful in determining **H**; that is,

$$H_\phi\oint_L dl = I_{enc} \quad \text{or} \quad H_\phi = \frac{I_{enc}}{\ell}$$

agnetic flux through a surface S is given by

$$\Psi = \int_S \mathbf{B}\cdot d\mathbf{S} \quad \text{(in Wb)}$$

B is the magnetic flux density (in Wb/m²). In free space,

$$\mathbf{B} = \mu_o\mathbf{H}$$

$\mu_o = 4\pi\times10^{-7}$ H/m = permeability of free space.
an isolated or free magnetic monopole does not exist, the net magnetic flux gh a closed surface is zero:

$$\Psi = \oint_S \mathbf{B}\cdot d\mathbf{S} = 0$$

$\nabla\cdot\mathbf{B} = 0$ (fourth Maxwell equation to be derived)

s point, all four Maxwell equations for static EM fields have been derived, namely:

$$\nabla\cdot\mathbf{D} = \rho_v$$
$$\nabla\cdot\mathbf{B} = 0$$
$$\nabla\times\mathbf{E} = \mathbf{0}$$
$$\nabla\times\mathbf{H} = \mathbf{J}$$

agnetic scalar potential V_m is defined as

$$\mathbf{H} = -\nabla V_m, \quad \text{if } \mathbf{J} = \mathbf{0}$$

ie magnetic vector potential **A** as

$$\mathbf{B} = \nabla\times\mathbf{A}$$

$\nabla\cdot\mathbf{A} = 0$. With the definition of **A**, the magnetic flux through a surface S can nd from

$$\Psi = \oint_L \mathbf{A}\cdot d\mathbf{l}$$

$$\mathbf{A} = \int \frac{\mu_o I\,d\mathbf{l}}{4\pi R}, \quad R = |\mathbf{r}-\mathbf{r}'|$$

een electric and magnetic fields exist. Som onding to Poisson's equation $\nabla^2 V = -\rho_v/\varepsilon$

$$\nabla^2\mathbf{A} = -\mu_o\mathbf{J}$$

Chapter 8

$$P = \frac{F}{S} = \frac{1}{2}BH = \frac{B^2}{2\mu_o}$$

$$\mathbf{F} = Q(\mathbf{E}+\mathbf{u}\times\mathbf{B}) = m\frac{d\mathbf{u}}{dt}$$

he force acting on a particle with charge Q in the presence of E the fundamental law relating EM to mechanics.

n the Lorentz force law, the force experienced by a current elen ic field **B** is

$$d\mathbf{F} = I\,d\mathbf{l}\times\mathbf{B}$$

iis, the magnetic field **B** is defined as the force per unit current

que on a current loop with magnetic moment **m** in a uniform n

$$\mathbf{T} = \mathbf{m}\times\mathbf{B} = IS\mathbf{a}_n\times\mathbf{B}$$

ietic dipole is a bar magnet or a small filamental current loop; it its **B** field lines are similar to the **E** field lines of an electric dip

material is subjected to a magnetic field, it becomes magnetize on **M** is the magnetic dipole moment per unit volume of the ma l,

$$\mathbf{M} = \chi_m\mathbf{H}$$

χ_m is the magnetic susceptibility of the material.

s of their magnetic properties, materials are either linear (diam ic) or nonlinear (ferromagnetic). For linear materials,

$$\mathbf{B} = \mu\mathbf{H} = \mu_o\mu_r\mathbf{H} = \mu_o(1+\chi_m)\mathbf{H} = \mu_o(\mathbf{H}+\mathbf{M})$$

μ = permeability and $\mu_r = \mu/\mu_o$ = relative permeability of th

$$\mathbf{B}_{1n} = \mathbf{B}_{2n}$$
$$(\mathbf{H}_1 - \mathbf{H}_2)\times\mathbf{a}_{n12} = \mathbf{K} \ \ or\ \ \mathbf{H}_{1t} = \mathbf{H}_{2t}, \ \text{if } \mathbf{K} = \mathbf{0},$$

where $\mathbf{a}_{n12}$ is a unit vector directed from medium 1 to medium 2.

Energy in a magnetostatic field is given by

$$W_m = \frac{1}{2}\int_v \mathbf{B}\cdot\mathbf{H}\,dv$$

For an inductor carrying current I

$$W_m = \frac{1}{2}LI^2$$

Thus the inductance L can be found by using

$$L = \frac{\int_v \mathbf{B}\cdot\mathbf{H}\,dv}{I^2}$$

The inductance L of an inductor can also be determined from its basic ratio of the magnetic flux linkage to the current through the inductor,

$$L = \frac{\lambda}{I} = \frac{N\Psi}{I}$$

Thus by assuming current I, we determine **B** and $\Psi = \int_s \mathbf{B}\cdot d\mathbf{S}$, and fi $L = N\Psi/I$.

A magnetic circuit can be analyzed in the same way as an electric circ keep in mind the similarity between

$$\mathcal{F} = NI = \oint \mathbf{H}\cdot d\mathbf{l} = \Psi\mathcal{R} \quad \text{and} \quad V = IR$$

that is,

$$\mathcal{F}\leftrightarrow V, \Psi\leftrightarrow I, \mathcal{R}\leftrightarrow R$$

Chapter 8

8.1 A 4 mC charge has velocity $\mathbf{u} = 1.4\mathbf{a}_x - 3.2\mathbf{a}_y - \mathbf{a}_z$ m/s at point $P(2, 5, -3)$ in the presence of $\mathbf{E} = 2xyz\mathbf{a}_x + x^2z\mathbf{a}_y + x^2y\mathbf{a}_z$ V/m and $\mathbf{B} = y^2\mathbf{a}_x + z^2\mathbf{a}_y + x^2\mathbf{a}_z$ Wb/m². Find the force on the charge at P.

8.2 An electron ($m = 9.11 \times 10^{-31}$ kg) moves in a circular orbit of radius 0.4×10^{-10} m with an angular velocity of 2×10^{16} rad/s. Find the centripetal force required to hold the electron.

8.3 A 1 mC charge with velocity $10\mathbf{a}_x - 2\mathbf{a}_y + 6\mathbf{a}_z$ m/s enters a region where the magnetic flux density is $25\mathbf{a}_z$ Wb/m². (a) Calculate the force on the charge. (b) Determine the electric field intensity necessary to make the velocity of the charge constant.

8.4 Assume an electric field intensity of 20 kV/m and a magnetic flux density of 5 μWb/m² exist in a region. Find the ratio of the magnitudes of electric and magnetic forces on an electron that has attained a velocity of 0.5×10^8 m/s.

8.5 A -2 mC charge starts at point $(0, 1, 2)$ with a velocity of $5\mathbf{a}_x$ m/s in a magnetic field $\mathbf{B} = 6\mathbf{a}_y$ Wb/m². Determine the position and velocity of the particle after 10 s, assuming that the mass of the charge is 1 gram. Describe the motion of the charge.

8.6 By injecting an electron beam normally to the plane edge of a uniform field $B_o\mathbf{a}_z$, electrons can be dispersed according to their velocity as in Figure 8.34.

(a) Show that the electrons would be ejected out of the field in paths parallel to the input beam as shown.

(b) Derive an expression for the exit distance d above the entry point.

8.7 Two large conducting plates are 8 cm apart and have a potential difference 12 kV. A drop of oil with mass 0.4 g is suspended in space between the plates. Find the charge on the drop.

8.8 A straight conductor 0.2 m long carries a current 4.5 A along $\mathbf{a}_x$. If the conductor lies in the magnetic field $\mathbf{B} = 2.5(\mathbf{a}_y + \mathbf{a}_z)$ mWb/m², calculate the force on the conductor.

8.9 Determine $|\mathbf{B}|$ that will produce the same force on a charged particle moving at 140 m/s that an electric field of 12 kV/m produces.

8.18 A 60-turn coil carries a current of 2 A and lies in the plane $x + 2y - 5z = 12$ such that the magnetic moment $\mathbf{m}$ of the coil is directed away from the origin. Calculate $\mathbf{m}$, assuming that the area of the coil is 8 cm².

8.19 The earth has a magnetic moment of about 8×10^{22} A · m² and its radius is 6370 km. Imagine that there is a loop around the equator and determine how much current in the loop would result in the same magnetic moment.

8.20 A triangular loop is placed in the x-z plane, as shown in Figure 8.39. Assume that a dc current $I = 2$ A flows in the loop and that $\mathbf{B} = 30\mathbf{a}_z$ m Wb/m exists in the region. Find the forces and torque on the loop.

8.21 A loop with 50 turns and surface area of 12 cm² carries a current of 4 A. If the loop rotates in a uniform magnetic field of 100 mWb/m², find the torque exerted on the loop.

8.22 High-current circuit breakers typically consist of coils that generate a magnetic field to blow out the arc formed when the contacts open. An arc 30 mm long carries a current of 520 A in a direction perpendicular to a magnetic flux density of 0.4 mWb/m². Determine the magnetic force on the arc.

8.24 A block of iron ($\mu = 5000\mu_o$) is placed in a uniform magnetic field with 1.5 Wb/m². If iron consists of 8.5×10^{28} atoms/m³, calculate (a) the magnetization M, (b) the average magnetic moment.

8.25 In a magnetic material, with $\chi_m = 6.5$, the magnetization is $\mathbf{M} = 24y^2\mathbf{a}_z$ A/m. Find μ_r, $\mathbf{H}$, and $\mathbf{J}$ at $y = 2$ cm.

8.26 In a ferromagnetic material ($\mu = 80\mu_o$), $\boldsymbol{B} = 20x\boldsymbol{a}_y$ mWb/m². Determine: (a) μ_r, (b) χ_m, (c) $\boldsymbol{H}$, (d) $\boldsymbol{M}$, (e) $\mathbf{J}_b$.

8.30 Region 1, for which $\mu_1 = 2.5\mu_o$, is defined by $z < 0$, while region 2, for which $\mu_2 = 4\mu_o$, is defined by $z > 0$. If $\mathbf{B}_1 = 6\mathbf{a}_x - 4.2\mathbf{a}_y + 1.8\mathbf{a}_z$ mWb/m², find $\mathbf{H}_2$ and the angle $\mathbf{H}_2$ makes with the interface.

8.31 In medium 1 ($z < 0$) $\mu_1 = 5\mu_o$, while in medium 2 ($z > 0$) $\mu_2 = 2\mu_o$. If $\boldsymbol{B}_1 = 4\boldsymbol{a}_x - 10\boldsymbol{a}_y + 12\boldsymbol{a}_z$ mWb/m², find B_2 and the energy density in medium 2.

8.32 In region $x < 0$, $\mu = \mu_o$, a uniform magnetic field makes angle 42° with the normal to the interface. Calculate the angle the field makes with the normal in region $x > 0$, $\mu = 6.5\mu_o$.

8.33 A current sheet with $\mathbf{K} = 12\mathbf{a}_y$ A/m is placed at $x = 0$, which separates region 1, $x < 0$, $\mu = 2\mu_o$ and region 2, $x > 0$, $\mu = 4\mu_o$. If $\mathbf{H}_1 = 10\mathbf{a}_x + 6\mathbf{a}_z$, A/m, find $\mathbf{H}_2$.

8.34 Suppose space is divided into region 1 ($y < 0$, $\mu_1 = \mu_o\mu_{r1}$) and region 2 ($y < 0$, $\mu_2 = \mu_o\mu_{r2}$). If $\mathbf{H}_1 = \alpha\mathbf{a}_x + \beta\mathbf{a}_y + \delta\mathbf{a}_z$ A/m, find $\mathbf{H}_2$.

8.42 An air-filled toroid of square cross section has inner radius 3 cm, outer radius 5 cm, and height 2 cm. How many turns are required to produce an inductance of 45 μH?

8.43 A wire of radius 2 mm is 40 m long. Calculate its inductance. Assume $\mu = \mu_o$.

8.44 A coaxial cable has an internal inductance that is twice the external inductance. If the inner radius is 6.5 mm, calculate the outer radius.

8.45 A hollow cylinder of radius $a = 2$ cm is 10 m long. Find the inductance of the cylinder. (See Table 8.3.)

8.46 Show that the mutual inductance between the rectangular loop and the infinite line current of Figure 8.4 is

$$M_{12} = \frac{\mu b}{2\pi} \ln\left[\frac{a + \rho_o}{\rho_o}\right]$$

Calculate M_{12} when $a = b = \rho_o = 1$ m.

8.50 In a certain region for which $\chi_m = 19$,

$$\mathbf{H} = 5x^2yz\mathbf{a}_x + 10xy^2z\mathbf{a}_y - 15xyz^2\mathbf{a}_z \text{ A/m}$$

How much energy is stored in $0 < x < 1, 0 < y < 2, -1 < z < 2$?

8.51 The magnetic field in a material space ($\mu = 15\mu_o$) is given by

$$\mathbf{B} = 4\mathbf{a}_x + 12\mathbf{a}_y \text{ mWb/m}^2$$

Calculate the energy stored in region $0 < x < 2, 0 < y < 3, 0 < z < 4$.

Chapter 7

7.1 (a) State Biot–Savart's law.

(b) The y- and z-axes, respectively, carry filamentary currents 10 A along $\mathbf{a}_y$ and 20 A along $-\mathbf{a}_z$. Find $\mathbf{H}$ at $(-3, 4, 5)$.

7.2 A long, straight wire carries current 2A. Calculate the distance from the wire when the magnetic field strength is 10 mA/m.

7.3 Two infinitely long wires, placed parallel to the z-axis, carry currents 10 A in opposite directions as shown in Figure 7.28. Find $\mathbf{H}$ at point P.

7.4 Two current elements $I_1dl_1 = 4 \times 10^{-5}\,\mathbf{a}_x$ A.m at (0, 0, 0) and $I_2dl_2 = 6 \times 10^{-5}\,\mathbf{a}_y$ A.m at (0, 0, 1) are in free space. Find $\mathbf{H}$ at (3, 1, –2).

7.6 Consider AB in Figure 7.29 as part of an electric circuit. Find $\mathbf{H}$ at the origin due to AB.

7.7 Line $x = 0$, $y = 0$, $0 \le z \le 10$ m carries current 2 A along $\mathbf{a}_z$. Calculate $\mathbf{H}$ at points

(a) $(5, 0, 0)$
(b) $(5, 5, 0)$
(c) $(5, 15, 0)$
(d) $(5, -15, 0)$

7.10 Find $\mathbf{H}$ at the center C of an equilateral triangular loop of side 4 m carrying 5 A of current as in Figure 7.31.

7.16 Two identical loops are parallel and separated by distance d as shown in Figure 7.35. Calculate $\mathbf{H}$ at $(0, 0, d)$ assuming that $a = 3$ cm, $d = 4$ cm, and $I = 10$ A.

7.17 A solenoid of radius 4 mm and length 2 cm has 150 turns/m and carries a current of 500 mA. Find (a) $|\mathbf{H}|$ at the center, (b) $|\mathbf{H}|$ at the ends of the solenoid.

7.34 In free space, the magnetic flux density is

$$\mathbf{B} = y^2\mathbf{a}_x + z^2\mathbf{a}_y + x^2\mathbf{a}_z \text{ Wb/m}^2$$

(a) Show that $\mathbf{B}$ is a magnetic field
(b) Find the magnetic flux through $x = 1, 0 < y < 1, 1 < z < 4$.
(c) Calculate $\mathbf{J}$.
(d) Determine the total magnetic flux through the surface of a cube defined by $0 < x < 2, 0 < y < 2, 0 < z < 2$.

7.37 In free space, $\mathbf{B} = \dfrac{20}{\rho}\sin^2\phi\,\mathbf{a}_z$ Wb/m². Determine the magnetic flux crossing the strip $z = 0, 1 < \rho < 2$ m, $0 < \phi < \pi/4$.

7.40 Consider the following arbitrary fields. Find out which of them can possibly represent an electrostatic or magnetostatic field in free space.

(a) $\mathbf{A} = y\cos ax\mathbf{a}_x + (y + e^{-x})\mathbf{a}_z$
(b) $\mathbf{B} = \dfrac{20}{\rho}\mathbf{a}_\rho$
(c) $\mathbf{C} = r^2\sin\theta\,\mathbf{a}_\phi$

7.41 Reconsider Problem 7.40 for the following fields.

(a) $\mathbf{D} = y^2z\mathbf{a}_x + 2(x + 1)yz\mathbf{a}_y - (x + 1)z^2\mathbf{a}_z$
(b) $\mathbf{E} = \dfrac{(z + 1)}{\rho}\cos\phi\,\mathbf{a}_\rho + \dfrac{\sin\phi}{\rho}\mathbf{a}_z$
(c) $\mathbf{F} = \dfrac{1}{r^2}(2\cos\theta\,\mathbf{a}_r + \sin\theta\,\mathbf{a}_\theta)$

Chapter 6

6.13 Two conducting coaxial cylinders are located at $\rho = 1$ cm and $\rho = 1.5$ cm. The inner conductor is maintained at 50 V while the outer one is grounded. If the cylinders are separated by a dielectric material with $\varepsilon = 4\varepsilon_o$, find the surface charge density on the inner conductor.

6.14 Consider the conducting plates shown in Figure 6.30. If $V(z = 0) = 0$ and $V(z = 2\text{ mm}) = 50$ V, determine V, $\mathbf{E}$, and $\mathbf{D}$ in the dielectric region ($\varepsilon_r = 1.5$) between the plates and ρ_S on the plates.

6.15 The cylindrical structure whose cross section is in Figure 6.31 has inner and outer radii of 5 mm and 15 mm, respectively. If $V(\rho = 5\text{ mm}) = 100$ V and $V(\rho = 15\text{ mm}) = 0$ V, calculate V, $\mathbf{E}$, and $\mathbf{D}$ at $\rho = 10$ mm and ρ_S on each plate. Take $\varepsilon_r = 2.0$.

6.16 In cylindrical coordinates, $V = 50$ V on plane $\phi = \pi/2$ and $V = 0$ on plane $\phi = 0$. Assuming that the planes are insulated along the z-axis, determine $\mathbf{E}$ between the planes.

6.29 Show that the resistance of the bar of Figure 6.17 between the vertical ends located at $\phi = 0$ and $\phi = \pi/2$ is

$$R = \frac{\pi}{2\sigma t \ln\dfrac{b}{a}}$$

'6.30 Show that the resistance of the sector of a spherical shell of conductivity σ, with cross section shown in Figure 6.37 (where $0 \le \phi < 2\pi$), between its base (i.e., from $r = a$ to $r = b$) is

$$R = \frac{1}{2\pi\sigma(1 - \cos\alpha)}\left[\frac{1}{a} - \frac{1}{b}\right]$$

6.31 A spherical shell has inner and outer radii a and b, respectively. Assume that the shell has a uniform conductivity σ and that it has copper electrodes plated on the inner and outer surfaces. Show that

$$R = \frac{1}{4\pi\sigma}\left(\frac{1}{a} - \frac{1}{b}\right)$$

6.66 Two point charges of 3 nC and -4 nC are placed, respectively, at $(0, 0, 1\text{ m})$ and $(0, 0, 2\text{ m})$ while an infinite conducting plane is at $z = 0$. Determine

(a) The total charge induced on the plane
(b) The magnitude of the force of attraction between the charges and the plane

add 7.37...

Prob. 8.1
At P, x = 2, y = 5, z = -3

$\boldsymbol{E}=2(2)(5)(-3)\boldsymbol{a}_x+(2)^2(-3)\boldsymbol{a}_y+(2)^2(5)\boldsymbol{a}_z=-60\boldsymbol{a}_x-12\boldsymbol{a}_y+20\boldsymbol{a}_z$

$\boldsymbol{B}=(5)^2\boldsymbol{a}_x+(-3)^2\boldsymbol{a}_y+2^2\boldsymbol{a}_z=25\boldsymbol{a}_x+9\boldsymbol{a}_y+4\boldsymbol{a}_z$

$\boldsymbol{F}=Q(\boldsymbol{E}+\boldsymbol{u}\times\boldsymbol{B})$

$\boldsymbol{u}\times\boldsymbol{B}=\begin{vmatrix}1.4 & -3.2 & -1\\ 25 & 9 & 4\end{vmatrix}=-3.8\boldsymbol{a}_x-30.6\boldsymbol{a}_y+92.6\boldsymbol{a}_z$

$\boldsymbol{E}+\boldsymbol{u}\times\boldsymbol{B}=(-60,-12,20)+(-3.8,-30.6,92.6)=(-63.8,-42.6,112.6)$

$\boldsymbol{F}=Q(\boldsymbol{E}+\boldsymbol{u}\times\boldsymbol{B})=4(\boldsymbol{E}+\boldsymbol{u}\times\boldsymbol{B})$ mN

$=\underline{\underline{-255.2\boldsymbol{a}_x-170.4\boldsymbol{a}_y+450.4\boldsymbol{a}_z\text{ mN}}}$

Prob. 8.2

$$F=m\omega^2r=9.11\times10^{-31}\times(2\times10^{16})^2(0.4\times10^{-10})=\underline{\underline{14.576\text{ nN}}}$$

Prob. 8.3
(a)

$\boldsymbol{F}=Q(\boldsymbol{u}\times\boldsymbol{B})=10^{-3}\begin{vmatrix}10 & -2 & 6\\ 0 & 0 & 25\end{vmatrix}=10^{-3}(-50\boldsymbol{a}_x-250\boldsymbol{a}_y)$

$=\underline{\underline{-0.05\boldsymbol{a}_x-0.25\boldsymbol{a}_y\text{ N}}}$

(b) Constant velocity implies that acceleration **a** = **0**.
$\boldsymbol{F}=m\boldsymbol{a}=0=Q(\boldsymbol{E}+\boldsymbol{u}\times\boldsymbol{B})$
$\boldsymbol{E}=-\boldsymbol{u}\times\boldsymbol{B}=\underline{\underline{50\boldsymbol{a}_x+250\boldsymbol{a}_y\text{ V/m}}}$

Prob. 8.4

$F_e=qE,\quad F_m=q\boldsymbol{u}\times\boldsymbol{B}$

$\dfrac{F_e}{F_m}=\dfrac{E}{uB}=\dfrac{20\times10^3}{0.5\times10^8\times5\times10^{-6}}=\underline{\underline{80}}$

Prob. 8.5
$m\boldsymbol{a}=Q\boldsymbol{u}\times\boldsymbol{B}$

$10^{-3}\boldsymbol{a}=-2\times10^{-3}\begin{vmatrix}u_x & u_y & u_z\\ 0 & 6 & 0\end{vmatrix}$

$\dfrac{d}{dt}(u_x,u_y,u_z)=(12u_z,0,-12u_x)$

i.e. $\dfrac{du_x}{dt}=12u_z$ (1

$\dfrac{du_y}{dt}=0\to u_y=A_1$ (2

$\dfrac{du_z}{dt}=-12u_x$ (3

From (1) and (3),

$\ddot{u}_x=12\dot{u}_z=-144u_x$

or

$\ddot{u}_x+144u_x=0\to u_x=c_1\cos12t+c_2\sin12t$

From (1), $u_z=-c_1\sin12t+c_2\cos12t$

At t=0,
$u_x=5,\ u_y=0,\ u_z=0\to A_1=0=c_2,\ c_1=5$

Hence,
$\boldsymbol{u}=(5\cos12t,0,-5\sin12t)$

$\boldsymbol{u}(t=10s)=(5\cos120,0,-5\sin120)=\underline{\underline{4.071\boldsymbol{a}_x-2.903\boldsymbol{a}_z\text{ m/s}}}$

$u_x=\dfrac{dx}{dt}=5\cos12t\to x=\frac{5}{12}\sin12t+B_1$

$u_y=\dfrac{dy}{dt}=0\to y=B_2$

$u_z=\dfrac{dz}{dt}=-5\sin12t\to z=\frac{5}{12}\cos12t+B_3$

At t=0, (x, y, z) = (0, 1, 2) → $B_1=0$, $B_2=1$, $B_3=\frac{19}{12}$

$(x,y,z)=\left(\dfrac{5}{12}\sin12t,1,\dfrac{5}{12}\cos12t+\dfrac{19}{12}\right)$ (4)

At t=10s,
$(x,y,z)=\left(\dfrac{5}{12}\sin120,1,\dfrac{5}{12}\cos120+\dfrac{19}{12}\right)=\underline{(0.2419,\ 1,\ 1.923)}$

By eliminating t from (4),
$x^2+(z-\frac{19}{12})^2=(\frac{5}{12})^2$, $y=1$ which is a circle in the y=1 plane with center at (0,1,19/12). The particle gyrates.

Prob. 8.6
(a) $m\boldsymbol{a}=-e(\boldsymbol{u}\times\boldsymbol{B})$

$-\dfrac{m}{e}\dfrac{d}{dt}(u_x,u_y,u_z)=\begin{vmatrix}u_x & u_y & u_z\\ 0 & 0 & B_o\end{vmatrix}=u_yB_o\vec{a}_x-B_ou_x\vec{a}_y$

$\dfrac{du_z}{dt}=0\to u_z=c=0$

$\dfrac{du_x}{dt}=-u_y\dfrac{B_oe}{m}=-u_yw$, where $w=\dfrac{B_oe}{m}$

$\dfrac{du_y}{dt}=u_xw$

Hence,

$\ddot{u}_x=-w\dot{u}_y=-w^2u_x$
or $\ddot{u}_x+w^2u_x=0\to u_x=A\cos wt+B\sin wt$

$u_y=-\dfrac{\dot{u}_x}{w}=A\sin wt-B\cos wt$

At t=0, $u_x=u_o$, $u_y=0\to A=u_o$, B=0

Hence,
$u_x=u_o\cos wt=\dfrac{dx}{dt}\to x=\dfrac{u_o}{w}\sin wt+c_1$

$u_y=u_o\sin wt=\dfrac{dy}{dt}\to y=-\dfrac{u_o}{w}\cos wt+c_2$

At t=0, x = 0 = y → $c_1=0$, $c_2=\frac{u_o}{w}$. Hence,

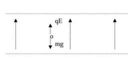
$x=\dfrac{u_o}{w}\sin wt,\ y=\dfrac{u_o}{w}(1-\cos wt)$

$\dfrac{u_o^2}{w^2}(\cos^2wt+\sin^2wt)=\left(\dfrac{u_o}{w}\right)^2=x^2+(y-\dfrac{u_o}{w})^2$

showing that the electron would move in a circle centered at $(0,\frac{u_o}{w})$. But since the field does not exist throughout the circular region, the electron passes through a semi-circle and leaves the field horizontally.

(b) d = twice the radius of the semi-circle
$=\dfrac{2u_o}{w}=\dfrac{2u_om}{B_oe}$

Prob. 8.7



$mg=qE\longrightarrow q=\dfrac{mg}{E}=\dfrac{0.4\times10^{-3}\times9.81}{1.5\times10^5}=\underline{\underline{26.67\text{ nC}}}$

Prob. 8.8
$\boldsymbol{F}=\int Id\boldsymbol{l}\times\boldsymbol{B}=I\boldsymbol{L}\times\boldsymbol{B}$

$=4.5(0.2)\boldsymbol{a}_x\times(2.5)(\boldsymbol{a}_y+\boldsymbol{a}_z)10^{-3}=4.5(0.5)\begin{vmatrix}\boldsymbol{a}_x & \boldsymbol{a}_y & \boldsymbol{a}_z\\ 1 & 0 & 0\\ 0 & 1 & 1\end{vmatrix}$ mN

$\boldsymbol{F}=\underline{2.25(-\boldsymbol{a}_y+\boldsymbol{a}_z)\text{ mN}}$

Prob. 8.9

$qE=quB\to B=\dfrac{E}{u}=\dfrac{12\times10^3}{140}=\dfrac{1200}{14}=\underline{\underline{85.714\text{ Wb/m}^2}}$

Prob. 8.18
$f(x,y,z)=x+2y-5z-12=0\longrightarrow\nabla f=\boldsymbol{a}_x+2\boldsymbol{a}_y-5\boldsymbol{a}_z$

$\boldsymbol{a}_n=\dfrac{\nabla f}{|\nabla f|}=\dfrac{\boldsymbol{a}_x+2\boldsymbol{a}_y-5\boldsymbol{a}_z}{\sqrt{30}}$

$\boldsymbol{m}=NIS\boldsymbol{a}_n=2\times60\times8\times10^{-4}\dfrac{(\boldsymbol{a}_x+2\boldsymbol{a}_y-5\boldsymbol{a}_z)}{\sqrt{30}}=\underline{\underline{17.53\boldsymbol{a}_x+35.05\boldsymbol{a}_y-87.64\boldsymbol{a}_z\text{ mAm}}}$

Prob. 8.19
$m=IS\longrightarrow I=\dfrac{m}{S}=\dfrac{m}{\pi r^2}$

$I=\dfrac{8\times10^{22}}{\pi(6370\times10^3)^2}=6.275\times10^8=\underline{\underline{627.5\text{ MA}}}$

Prob. 8.20

Let $\boldsymbol{F}=\boldsymbol{F}_1+\boldsymbol{F}_2+\boldsymbol{F}_3$

$\boldsymbol{F}_1=\int Id\boldsymbol{l}\times\boldsymbol{B}=\int_5^0 2dx\boldsymbol{a}_x\times30\boldsymbol{a}_z$ mN

$=-60\boldsymbol{a}_yx\Big|_5^0=300\boldsymbol{a}_y$ mN

$\boldsymbol{F}_2=\int_0^5 2dy\boldsymbol{a}_y\times30\boldsymbol{a}_z$ mN

$=60\boldsymbol{a}_xy\Big|_0^5=300\boldsymbol{a}_x$ mN

$\boldsymbol{F}_3=\int_0^5 2(dx\boldsymbol{a}_x+dz\boldsymbol{a}_z)\times30\boldsymbol{a}_z$ mN

$=60(-\boldsymbol{a}_y)x\Big|_0^5=-300\boldsymbol{a}_y$ mN

$\boldsymbol{F}=\boldsymbol{F}_1+\boldsymbol{F}_2+\boldsymbol{F}_3=300\boldsymbol{a}_y+300\boldsymbol{a}_x-300\boldsymbol{a}_y$ mN $=\underline{\underline{300\boldsymbol{a}_x\text{ mN}}}$

$\boldsymbol{T}=\boldsymbol{m}\times\boldsymbol{B}=IS\boldsymbol{a}_n\times\boldsymbol{B}=2(\frac{1}{2})(5)(5)\boldsymbol{a}_y\times30\boldsymbol{a}_z10^{-3}=\underline{\underline{0.75\boldsymbol{a}_x\text{ N.m}}}$

Prob. 8.21
For each turn, $\boldsymbol{T}=\boldsymbol{m}\times\boldsymbol{B}$, $\boldsymbol{m}=IS\boldsymbol{a}_n$
For N turns,
$T=NISB=50\times4\times12\times10^{-4}\times100\times10^{-3}=\underline{\underline{24\text{ mNm}}}$

Prob. 8.24

(a) $\boldsymbol{M}=\chi_m\boldsymbol{H}=\chi_m\dfrac{\boldsymbol{B}}{\mu_o\mu}$

$M=\dfrac{4999}{5000}\times\dfrac{1.5}{4\pi\times10^{-7}}=\underline{\underline{1.193\times10^6\text{ A/m}}}$

(b) $\boldsymbol{M}=\dfrac{\sum_{k=1}^{N}\boldsymbol{m}_k}{\Delta v}$

If we assume that all $\boldsymbol{m}_k$ align with the applied **B** field,

$M=\dfrac{Nm_k}{\Delta v}\to m_k=\dfrac{M}{N/\Delta v}=\dfrac{1.193\times10^6}{8.5\times10^{28}}$

$m_k=\underline{\underline{1.404\times10^{-23}\text{ A}\cdot\text{m}^2}}$

Prob. 8.25

$\mu_r=\chi_m+1=6.5+1=\underline{\underline{7.5}}$

$\boldsymbol{M}=\chi_m\boldsymbol{H}\longrightarrow\boldsymbol{H}=\dfrac{\boldsymbol{M}}{\chi_m}=\dfrac{24y^2}{6.5}\boldsymbol{a}_z$

At y = 2cm,

$\boldsymbol{H}=\dfrac{24\times4\times10^{-4}}{6.5}\boldsymbol{a}_z=\underline{\underline{1.477\boldsymbol{a}_z\text{ mA/m}}}$

$\boldsymbol{J}=\nabla\times\boldsymbol{H}=\begin{vmatrix}\dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ 0 & 0 & \dfrac{24y^2}{6.5}\end{vmatrix}=\dfrac{48y}{6.5}\mathbf{a}_x$

At y=2cm,

$\boldsymbol{J}=\dfrac{48\times2\times10^{-2}}{6.5}\mathbf{a}_x=\underline{\underline{0.1477\boldsymbol{a}_x\text{ A/m}^2}}$

Prob. 8.26

(a) $\mu=80\mu_o\to\mu_r=80$
(b) $\chi_m=\mu_r-1=79$

(c) $\boldsymbol{H}=\dfrac{\boldsymbol{B}}{\mu}=\dfrac{20x\boldsymbol{a}_y10^{-3}}{80(4\pi\times10^{-7})}=198.9x\boldsymbol{a}_y$ A/m

(d) $\boldsymbol{M}=\chi_m\boldsymbol{H}=15.713x\boldsymbol{a}_y$ kA/m

(e) $\boldsymbol{J}_b=\nabla\times\boldsymbol{M}=\begin{vmatrix}\dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ 0 & 15.713x & 0\end{vmatrix}=15.713\boldsymbol{a}_z$ kA/m

Prob. 8.30

$\boldsymbol{B}_{2n}=\boldsymbol{B}_{1n}=1.8\boldsymbol{a}_z$

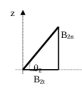
$\boldsymbol{H}_{2t}=\boldsymbol{H}_{1t}\longrightarrow\dfrac{\boldsymbol{B}_{2t}}{\mu_2}=\dfrac{\boldsymbol{B}_{1t}}{\mu_1}$

$\boldsymbol{B}_{2t}=\dfrac{\mu_2}{\mu_1}\boldsymbol{B}_{1t}=\dfrac{4\mu_o}{2.5\mu_o}(6\boldsymbol{a}_x-4.2\boldsymbol{a}_y)=9.6\boldsymbol{a}_x-6.72\boldsymbol{a}_y$

$\boldsymbol{B}_2=\boldsymbol{B}_{2n}+\boldsymbol{B}_{2t}=9.6\boldsymbol{a}_x-6.72\boldsymbol{a}_y+1.8\boldsymbol{a}_z$ mWb/m²

$\boldsymbol{H}_2=\dfrac{\boldsymbol{B}_2}{\mu_2}=\dfrac{10^{-3}(9.6,-6.72,1.8)}{4\times4\pi\times10^{-7}}$

$=\underline{\underline{1{,}909.86\,\boldsymbol{a}_x-1{,}336.9\boldsymbol{a}_y+358.1\boldsymbol{a}_z\text{ A/m}}}$



$\tan\theta_2=\dfrac{B_{2n}}{B_{2t}}=\dfrac{1.8}{\sqrt{9.6^2+6.72^2}}=0.1536$

$\theta_2=\underline{\underline{8.73^o}}$

Prob. 8.31

$\boldsymbol{B}_{2n}=\boldsymbol{B}_{1n}=12\boldsymbol{a}_z$

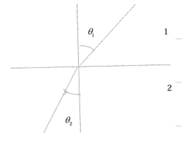
$\boldsymbol{H}_{2t}=\boldsymbol{H}_{1t}\longrightarrow\dfrac{\boldsymbol{B}_{2t}}{\mu_2}=\dfrac{\boldsymbol{B}_{1t}}{\mu_1}$

$\boldsymbol{B}_{2t}=\dfrac{\mu_2}{\mu_1}\boldsymbol{B}_{1t}=\dfrac{2\mu_o}{5\mu_o}(4\boldsymbol{a}_x-10\boldsymbol{a}_y)=1.6\boldsymbol{a}_x-4\boldsymbol{a}_y$

$\boldsymbol{B}_2=\boldsymbol{B}_{2n}+\boldsymbol{B}_{2t}=\underline{\underline{1.6\boldsymbol{a}_x-4\boldsymbol{a}_y+12\boldsymbol{a}_z\text{ mWb/m}^2}}$

$w_2=\dfrac{1}{2}\mu_2H_2^2=\dfrac{B_2^2}{2\mu_2}=\dfrac{(1.6^2+4^2+12^2)\times10^{-6}}{2(2)(4\pi\times10^{-7})}=\underline{\underline{32.34\text{ J/m}^3}}$

Prob. 8.32



$\tan\theta_1=\dfrac{H_{1t}}{H_{1n}},\qquad\tan\theta_2=\dfrac{H_{2t}}{H_{2n}}$

But $H_{1t}=H_{2t}$

$B_{1n}=B_{2t}\to\mu_1H_{1n}=\mu_2H_{2n}\to H_{2n}=\dfrac{\mu_1}{\mu_2}H_{1n}$

$\dfrac{H_{2t}}{H_{2n}}=\dfrac{H_{1t}}{\frac{1}{6.5}H_{1n}}=\dfrac{6.5H_{1t}}{H_{1n}}$

If $\theta_1=42^o$, $\tan42^o=\dfrac{H_{1t}}{H_{1n}}\to\dfrac{H_{1t}}{H_{1n}}=0.9004$

$\tan\theta_2=6.5(0.9004)=5.832\to\underline{\underline{\theta_2=80.3^o}}$

Prob. 8.33

x <0

x > 0

$(\boldsymbol{H}_1-\boldsymbol{H}_2)\times\boldsymbol{a}_{n12}=\boldsymbol{K}\to\boldsymbol{H}_1\times\mathbf{a}_x=\boldsymbol{K}+\boldsymbol{H}_2\times\boldsymbol{a}_x$
$\boldsymbol{H}_1\times\mathbf{a}_x=(10\boldsymbol{a}_x+6\boldsymbol{a}_z)\times\boldsymbol{a}_x=6\boldsymbol{a}_y$
$\boldsymbol{H}_2\times\mathbf{a}_x=(H_{2x},H_{2y},H_{2z})\times\boldsymbol{a}_x=H_{2z}\boldsymbol{a}_y-H_{2y}\boldsymbol{a}_z$
$6\boldsymbol{a}_y=12\boldsymbol{a}_y+H_{2z}\boldsymbol{a}_y-H_{2y}\boldsymbol{a}_z$
Equating components,
$6=12+H_{2z}\to H_{2z}=-6,$
$H_{2y}=0$
Also, $\boldsymbol{B}_{1n}=\boldsymbol{B}_{2n}\to\mu_1\boldsymbol{H}_{1n}=\mu_2\boldsymbol{H}_{2n}$
$\boldsymbol{H}_{2n}=\dfrac{\mu_1}{\mu_2}\boldsymbol{H}_{1n}=\dfrac{2\mu_o}{4\mu_o}(10\boldsymbol{a}_x)=5\boldsymbol{a}_x$
$\underline{\underline{\boldsymbol{H}_2=5\boldsymbol{a}_x-6\boldsymbol{a}_z\text{ A/m}}}$

Prob. 8.34

$\boldsymbol{H}_{2t}=\boldsymbol{H}_{1t}=\alpha\boldsymbol{a}_x+\delta\boldsymbol{a}_z$
$\boldsymbol{B}_{2n}=\boldsymbol{B}_{1n}\longrightarrow\mu_2\boldsymbol{H}_{2n}=\mu_1\boldsymbol{H}_{1n}$
$\boldsymbol{H}_{2n}=\dfrac{\mu_1}{\mu_2}\boldsymbol{H}_{1n}=\dfrac{\mu_{r1}}{\mu_{r2}}\beta\boldsymbol{a}_y$
$\underline{\underline{\boldsymbol{H}=\alpha\boldsymbol{a}_x+\dfrac{\mu_{r1}}{\mu_{r2}}\beta\boldsymbol{a}_y+\delta\boldsymbol{a}_z}}$

Prob. 8.42

$\rho_o=\dfrac{1}{2}(3+5)=4\text{cm}$
$a=2$ cm
$L=\dfrac{\mu_oN^2a}{2\pi}\ln\left[\dfrac{2\rho_o+a}{2\rho_o-a}\right]$

$N^2=\dfrac{2\pi L}{\mu_oa\ln\left[\dfrac{2\rho_o+a}{2\rho_o-a}\right]}=\dfrac{2\pi(45\times10^{-6})}{4\pi\times10^{-7}(2\times10^{-2})\ln\left(\dfrac{8+2}{8-2}\right)}=22{,}023.17$

$\underline{\underline{N=148.4\text{ or }148}}$

Prob. 8.43

$L=\dfrac{\mu_o\ell}{8\pi}=\dfrac{4\pi\times10^{-7}(40)}{8\pi}=20\times10^{-7}=\underline{\underline{2\ \mu\text{H}}}$

Prob. 8.44

$L_{in}=\dfrac{\mu_o\ell}{8\pi},\qquad L_{ext}=\dfrac{\mu_o\ell}{2\pi}\ln(b/a)$

If $L_{in}=2L_{ext}\longrightarrow\dfrac{\mu_o\ell}{8\pi}=\dfrac{\mu_o\ell}{\pi}\ln(b/a)$

$\ln(b/a)=\dfrac{1}{8}\qquad\dfrac{b}{a}=e^{1/8}=1.1331$

$b=1.1331a=\underline{7.365\text{ mm}}$

Prob. 8.45

$L=\dfrac{\mu_o\ell}{2\pi}\left[\ln\dfrac{2\ell}{a}-1\right]=\dfrac{4\pi\times10^{-7}(10)}{2\pi}\left[\ln\dfrac{2\times10}{2\times10^{-2}}-1\right]=2\times10^{-6}(\ln1000-1)$

$=2(5.908)\ \mu H=\underline{\underline{11.82\ \mu\text{H}}}$

Prob. 8.46

$\psi_{12}=\int\boldsymbol{B}_1\bullet d\boldsymbol{S}=\int_{\rho=\rho_o}^{\rho_o+a}\int_{z=0}^{b}\dfrac{\mu_oI}{2\pi\rho}dzd\rho=\dfrac{\mu_oIb}{2\pi}\ln\dfrac{a+\rho_o}{\rho_o}$

$M_{12}=\dfrac{N\psi_{12}}{I}=\dfrac{N\mu_ob}{2\pi}\ln\dfrac{a+\rho_o}{\rho_o}$

For N = 1,

$M_{12}=\dfrac{\psi_{12}}{I_1}=\dfrac{\mu_ob}{2\pi}\ln\dfrac{a+\rho_o}{\rho_o}$

$=\dfrac{4\pi\times10^{-7}}{2\pi}(1)\ln2=\underline{\underline{0.1386\mu\text{H}}}$

Prob. 8.50

$\mu_r=\chi_m+1=20$

$w_m=\dfrac{1}{2}\boldsymbol{B}_1\cdot\boldsymbol{H}_1=\dfrac{1}{2}\mu\mathbf{H}\cdot\mathbf{H}$

$=\dfrac{1}{2}\mu\left(25x^4y^2z^2+100x^2y^4z^2+225x^2y^2z^4\right)$

$W_m=\int w_mdv$

$=\dfrac{1}{2}\mu\left[25\int_0^1x^4dx\int_0^2y^2dy\int_{-1}^2z^2dz+100\int_0^1x^2dx\int_0^2y^4dy\int_{-1}^2z^2dz+225\int_0^1x^2dx\int_0^2y^2dy\int_{-1}^2zdz\right]$

$=\dfrac{25\mu}{2}\left[\dfrac{x^5}{5}\Big|_0^1\dfrac{y^3}{3}\Big|_0^2\dfrac{z^3}{3}\Big|_{-1}^2+4\dfrac{x^3}{3}\Big|_0^1\dfrac{y^5}{5}\Big|_0^2\dfrac{z^3}{3}\Big|_{-1}^2+9\dfrac{x^3}{3}\Big|_0^1\dfrac{y^3}{3}\Big|_0^2\dfrac{z^5}{5}\Big|_{-1}^2\right]$

$=\dfrac{25\mu}{2}\left(\dfrac{1}{5}\cdot\dfrac{8}{3}\cdot\dfrac{9}{3}+\dfrac{4}{3}\cdot\dfrac{32}{3}\cdot\dfrac{9}{3}+\dfrac{9}{3}\cdot\dfrac{8}{3}\cdot\dfrac{33}{5}\right)$

$=\dfrac{25}{2}\times4\pi\times10^{-7}\times20\times\dfrac{3600}{45}$

$W_m=\underline{\underline{25.13\text{ mJ}}}$

Prob. 8.51

$W=\dfrac{1}{2}\int_v\boldsymbol{B}\bullet\boldsymbol{H}dv=\int_v\dfrac{|\boldsymbol{B}|^2}{2\mu}dv=\dfrac{1}{2(15)4\pi\times10^{-7}}\int_{z=0}^{4}\int_{y=0}^{3}\int_{x=0}^{2}(4^2+12^2)10^{-6}dxdydz$

$=\dfrac{10^{-6}}{(30)4\pi\times10^{-7}}(16+144)(2)(3)(4)=\dfrac{320}{\pi}=\underline{\underline{101.86\text{ J}}}$

Prob. 7.1
(a) See text
(b) Let $\mathbf{H}=\mathbf{H}_y+\mathbf{H}_z$

For $\boldsymbol{H}_z=\dfrac{I}{2\pi\rho}\boldsymbol{a}_\phi\qquad\rho=\sqrt{(-3)^2+4^2}=5$

$\mathbf{a}_\phi=-\mathbf{a}_z\times\dfrac{(-3\mathbf{a}_x+4\mathbf{a}_y)}{5}=\dfrac{(3\mathbf{a}_y-4\mathbf{a}_x)}{5}$

$\mathbf{H}_z=\dfrac{20}{2\pi(25)}(4\mathbf{a}_x+3\mathbf{a}_y)=0.5093\mathbf{a}_x+0.382\mathbf{a}_y$

For $\mathbf{H}_y=\dfrac{\mathbf{I}}{2\pi\rho}\mathbf{a}_\phi$, $\rho=\sqrt{(-3)^2+5^2}=\sqrt{34}$

$\boldsymbol{a}_\phi=\boldsymbol{a}_y\times\dfrac{(-3\boldsymbol{a}_z+5\boldsymbol{a}_x)}{\sqrt{34}}=\dfrac{3\boldsymbol{a}_z+5\boldsymbol{a}_x}{\sqrt{34}}$

$H_y=\dfrac{10}{2\pi(34)}(5\boldsymbol{a}_x+3\boldsymbol{a}_z)=0.234\boldsymbol{a}_x+0.1404\boldsymbol{a}_z$

$\mathbf{H}=\mathbf{H}_y+\mathbf{H}_z$

$=\underline{0.7433\mathbf{a}_x+0.382\mathbf{a}_y+0.1404\mathbf{a}_z\text{ A/m}}$

Prob. 7.2

$H=\dfrac{I}{2\pi\rho}\to\rho=\dfrac{I}{2\pi H}=\dfrac{2}{2\pi(10\times10^{-3})}=\dfrac{100}{\pi}=\underline{\underline{31.83\text{ m}}}$

Prob. 7.3

Let $H=H_1+H_2$
where H_1 and H_2 are respectively due to the lines located at (0,0) and (0,5).

$\boldsymbol{H}_1=\dfrac{I}{2\pi\rho}\boldsymbol{a}_\phi,\qquad\rho=5,\qquad\boldsymbol{a}_\phi=\boldsymbol{a}_\ell\times\boldsymbol{a}_\rho=\boldsymbol{a}_z\times\boldsymbol{a}_x=\boldsymbol{a}_y$

$\boldsymbol{H}_1=\dfrac{10}{2\pi(5)}\boldsymbol{a}_y=\dfrac{\boldsymbol{a}_y}{\pi}$

$\boldsymbol{H}_2=\dfrac{I}{2\pi\rho}\boldsymbol{a}_\phi,\qquad\rho=5\sqrt{2},\qquad\boldsymbol{a}_\phi=\boldsymbol{a}_\ell\times\boldsymbol{a}_\rho,\ \boldsymbol{a}_\ell=-\boldsymbol{a}_z$

$\boldsymbol{a}_\rho=\dfrac{5\boldsymbol{a}_x-5\boldsymbol{a}_y}{5\sqrt{2}}=\dfrac{\boldsymbol{a}_x-\boldsymbol{a}_y}{\sqrt{2}}$

$\boldsymbol{a}_\phi=-\boldsymbol{a}_z\times\left(\dfrac{\boldsymbol{a}_x-\boldsymbol{a}_y}{\sqrt{2}}\right)=\dfrac{-\boldsymbol{a}_x-\boldsymbol{a}_y}{\sqrt{2}}$

$\boldsymbol{H}_2=\dfrac{10}{2\pi5\sqrt{2}}\left(\dfrac{-\boldsymbol{a}_x-\boldsymbol{a}_y}{\sqrt{2}}\right)=\dfrac{1}{2\pi}(-\boldsymbol{a}_x-\boldsymbol{a}_y)$

$\boldsymbol{H}=\boldsymbol{H}_1+\boldsymbol{H}_2=\dfrac{\boldsymbol{a}_y}{\pi}+\dfrac{1}{2\pi}(-\boldsymbol{a}_x-\boldsymbol{a}_y)=\underline{\underline{-0.1592\boldsymbol{a}_x+0.1592\boldsymbol{a}_y}}$

Prob. 7.10
For the side of the loop along y-axis,

$\boldsymbol{H}_1=\dfrac{I}{4\pi\rho}(\cos\alpha_2-\cos\alpha_1)\,\boldsymbol{a}_\phi$

where $\boldsymbol{a}_\phi=-\boldsymbol{a}_x$, $\rho=2\tan30^o=\dfrac{2}{\sqrt{3}}$, $\alpha_2=30^o$, $\alpha_1=150^o$

$\boldsymbol{H}_1=\dfrac{5}{4\pi}\dfrac{\sqrt{3}}{2}\left(\cos30^o-\cos150^o\right)(-\boldsymbol{a}_x)=-\dfrac{15}{8\pi}\boldsymbol{a}_x$

$\boldsymbol{H}=3\boldsymbol{H}_1=-1.79\boldsymbol{a}_x$ A/m

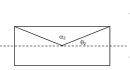
Prob. 7.16
From Example 7.3,

$\boldsymbol{H}_1=\dfrac{I}{2a}\boldsymbol{a}_z,\qquad\boldsymbol{H}_2=\dfrac{Ia^2}{2[a^2+d^2]^{3/2}}\boldsymbol{a}_z$

$\boldsymbol{H}=\boldsymbol{H}_1+\boldsymbol{H}_2=\dfrac{I}{2}\left[\dfrac{1}{a}+\dfrac{a^2}{[a^2+d^2]^{3/2}}\right]\boldsymbol{a}_z=\dfrac{10}{2}\left[\dfrac{1}{3\times10^{-2}}+\dfrac{3^2}{[3^2+4^2]^{3/2}(10^{-2})}\right]\boldsymbol{a}_z$

$=500\left[\dfrac{1}{3}+\dfrac{9}{125}\right]\boldsymbol{a}_z=\underline{202.67\boldsymbol{a}_z\text{ A/m}}$

Prob. 7.17

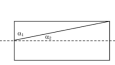


$|\boldsymbol{H}|=\dfrac{nI}{2}(\cos\theta_2-\cos\theta_1)$

$\cos\theta_2=-\cos\theta_1=\dfrac{\ell/2}{\left(a^2+\ell^2/4\right)^{1/2}}$

$|\boldsymbol{H}|=\dfrac{nI\ell}{2\left(a^2+\ell^2/4\right)^{1/2}}=\dfrac{0.5\times150\times2\times10^{-2}}{2\times10^{-3}\times\sqrt{4^2+10^2}}=\underline{69.63\text{ A/m}}$

(b)



$\alpha_1=90^o$, $\tan\theta_2=\dfrac{a}{b}=\dfrac{4}{20}=0.2\to\theta_2=11.31^o$

$|\boldsymbol{H}|=\dfrac{nI}{2}\cos\theta_2=\dfrac{150\times0.5}{2}\cos11.31^o=\underline{36.77\text{ A/m}}$

Prob. 7.34

(a) $\nabla\bullet\boldsymbol{B}=\dfrac{\partial B_x}{\partial x}+\dfrac{\partial B_y}{\partial y}+\dfrac{\partial B_z}{\partial z}=0$
showing that **B** satisfies Maxwell's equation.

(b) $d\boldsymbol{S}=dydz\boldsymbol{a}_x$

$\Psi=\int\boldsymbol{B}\bullet d\boldsymbol{S}=\int_{z=1}^{4}\int_{y=0}^{1}y^2dydz=\dfrac{y^3}{3}\Big|_0^1(z)\Big|_1^4=\underline{\underline{1\text{ Wb}}}$

(c) $\nabla\times\mathbf{H}=\mathbf{J}\longrightarrow\boldsymbol{J}=\nabla\times\dfrac{\boldsymbol{B}}{\mu_o}$

$\nabla\times\boldsymbol{B}=\begin{vmatrix}\dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ y^2 & z^2 & x^2\end{vmatrix}=-2z\boldsymbol{a}_x-2x\boldsymbol{a}_y-2y\boldsymbol{a}_z$

$\boldsymbol{J}=-\dfrac{2}{\mu_o}(z\boldsymbol{a}_x+x\boldsymbol{a}_y+y\boldsymbol{a}_z)$ A/m²

(d)
Since $\nabla\bullet\boldsymbol{B}=0$,
$\Psi=\oint_S\boldsymbol{B}\bullet d\boldsymbol{S}=\int_v\nabla\bullet\boldsymbol{B}dv=\underline{\underline{0}}$

Prob. 7.40

(a) $\nabla\cdot\boldsymbol{A}=-ya\sin ax\neq0$

$\nabla\times\boldsymbol{A}=\begin{vmatrix}\dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ y\cos ax & 0 & y+e^x\end{vmatrix}$

$=\boldsymbol{a}_x+e^{-x}\boldsymbol{a}_y-\cos ax\boldsymbol{a}_z\neq\boldsymbol{0}$

A is <u>neither</u> electrostatic nor magnetostatic field

(b) $\nabla\cdot\boldsymbol{B}=\dfrac{1}{\rho}\dfrac{\partial}{\partial\rho}(\rho B_\rho)=\dfrac{1}{\rho}\dfrac{\partial}{\partial\rho}(20)=0$

$\nabla\times\boldsymbol{B}=\boldsymbol{0}$

B <u>can be **E**-field in a charge-free region.</u>

(c) $\nabla\cdot\mathbf{C}=\dfrac{1}{r\sin\theta}\dfrac{\partial}{\partial\phi}(r^2\sin\theta)=0$

$\nabla\times\boldsymbol{C}=\dfrac{1}{r\sin\theta}\dfrac{\partial}{\partial\theta}\left(r^2\sin^2\theta\right)\boldsymbol{a}_r-\dfrac{1}{r}\dfrac{\partial}{\partial r}(r^3\sin\theta)\boldsymbol{a}_\theta\neq\boldsymbol{0}$

C is possibly <u>***H*** field.</u>

Prob. 7.41
(a) $\nabla\cdot\boldsymbol{D}=0$

$\nabla\times\boldsymbol{D}=\begin{vmatrix}\dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ y^2z & 2(x+1)yz & -(x+1)z^2\end{vmatrix}$

$=2(x+1)y\boldsymbol{a}_x+\ldots\neq\boldsymbol{0}$

D is possibly a <u>magnetostatic field.</u>

(b) $\nabla\cdot\mathbf{E}=\dfrac{1}{\rho}\dfrac{\partial}{\partial\rho}((z+1)\cos\phi)+\dfrac{\partial}{\partial z}\left(\dfrac{\sin\phi}{\rho}\right)=0$

$\nabla\times\boldsymbol{E}=\dfrac{1}{\rho^2}\cos\theta\,\boldsymbol{a}_\rho+\ldots\neq\boldsymbol{0}$

E could be a <u>magnetostatic</u> field.

(c) $\nabla\cdot\boldsymbol{F}=\dfrac{1}{r^2}\dfrac{\partial}{\partial r}(2\cos\theta)+\dfrac{1}{r\sin\theta}\dfrac{\partial}{\partial\theta}\left(\dfrac{\sin\theta}{r^2}\right)\neq0$

$\nabla\times\boldsymbol{F}=\dfrac{1}{r}\left[\dfrac{\partial}{\partial r}\left(r^{-1}\sin\theta\right)+\dfrac{2\sin\theta}{r^2}\right]\boldsymbol{a}_\theta\neq\boldsymbol{0}$

F can be <u>neither</u> electrostatic nor magnetostatic field.

Prob. 7.37

$\psi=\int_S\boldsymbol{B}\cdot d\boldsymbol{S}=\int_{\phi=0}^{\pi/4}\int_{\rho=1}^{2}\dfrac{20}{\rho}\sin^2\phi\,\rho d\rho d\phi=20\int_1^2d\rho\int_0^{\pi/4}\sin^2\phi d\phi$

$=20(1)\int_0^{\pi/4}\dfrac{1}{2}(1-\cos2\phi)\,d\phi=10(\phi-\dfrac{1}{2}\sin2\phi)\Big|_0^{\pi/4}$

$=10(\dfrac{\pi}{4}-\dfrac{1}{2})=\underline{\underline{2.854\text{ Wb}}}$

Prob. 6.13

$\nabla^2V=\dfrac{1}{\rho}\dfrac{d}{d\rho}\left(\rho\dfrac{dV}{d\rho}\right)=0\to V=A\ln\rho+B$

Let a = 1 cm, b = 1.5 cm, $V_o=50$V

$V(\rho=b)=0\to0=A\ln b+B$ or $B=-A\ln b$

$V(\rho=a)=V_o\to V_o=A\ln a-A\ln b=A\ln\dfrac{a}{b}$ or $A=\dfrac{V_o}{\ln\frac{a}{b}}$

$V=A\ln\rho-A\ln b=A\ln\dfrac{\rho}{b}$

$\boldsymbol{E}=-\nabla V=-\dfrac{dV}{d\rho}\boldsymbol{a}_\rho=-\dfrac{A}{\rho}\boldsymbol{a}_\rho=-\dfrac{V_o}{\rho\ln\frac{a}{b}}\boldsymbol{a}_\rho$

$\rho_s=D_n=\varepsilon E_n=\dfrac{\varepsilon_o\varepsilon_rV_o}{a\ln\frac{b}{a}}=\dfrac{50(4)\frac{10^{-9}}{36\pi}}{10^{-2}\ln1.5}=\dfrac{400(50)}{36\pi\ln1.5}\text{ nC/m}^2=\underline{\underline{436.14\text{ nC/m}^2}}$

Prob. 6.14

$\nabla^2V=\dfrac{d^2V}{dz^2}=0\longrightarrow V=Az+B$

When z=0, V = 0 ⟶ B=0

When z=d, V = V_o ⟶ $V_o=Ad$ or $A=V_o/d$

Hence,

$V=\dfrac{V_oz}{d}$

$\boldsymbol{E}=-\nabla V=-\dfrac{dV}{dz}\boldsymbol{a}_z=-\dfrac{V_o}{d}\boldsymbol{a}_z$

$\boldsymbol{D}=\varepsilon\boldsymbol{E}=-\varepsilon_o\varepsilon_r\dfrac{V_o}{d}\boldsymbol{a}_z$

Since $V_o=50$ V and d = 2mm,

<u>V = 25z kV,</u> <u>**E** = - 25**a**z kV/m</u>

$\boldsymbol{D}=-\dfrac{10^{-9}}{36\pi}(1.5)25\times10^3\boldsymbol{a}_z=\underline{\underline{-332\boldsymbol{a}_z\text{ nC/m}^2}}$

$\rho_s=D_n=\underline{\underline{\pm332\text{ nC/m}^2}}$

The surface charge density is positive on the plate at z=d and negative on the plate at z=0.

Prob. 6.15 From Example 6.8, solving $\nabla^2V=0$ when $V=V(\rho)$ leads to

$V=\dfrac{V_o\ln\rho/a}{\ln b/a}=V_o\dfrac{\ln(a/\rho)}{\ln(a/b)}$

$\mathbf{E}=-\nabla V=-\dfrac{V_o}{\rho\ln b/a}\mathbf{a}_\rho=\dfrac{V_o}{\rho\ln a/b}\mathbf{a}_\rho,\qquad\mathbf{D}=\varepsilon\mathbf{E}=-\dfrac{\varepsilon_o\varepsilon_rV_o}{\rho\ln b/a}\mathbf{a}_\rho$

$\rho_s=D_n=\pm\dfrac{\varepsilon_o\varepsilon_rV_o}{\rho\ln b/a}\Big|_{\rho=a,b}$

In this case, V_o=100 V, b=5mm, a=15mm, $\varepsilon_r=2$. Hence at $\rho=10$mm,

$V=\dfrac{100\ln(10/15)}{\ln(5/15)}=\underline{\underline{36.91\text{ V}}}$

$\mathbf{E}=\dfrac{100}{10\text{x}10^{-3}\ln3}\mathbf{a}_\rho=\underline{\underline{9.102\mathbf{a}_\rho\text{ kV/m}}}$

$\mathbf{D}=9.102\text{x}10^3\text{x}\dfrac{10^{-9}}{36\pi}2\mathbf{a}_\rho=\underline{\underline{161\mathbf{a}_\rho\text{ nC/m}^2}}$

$\rho_s(\rho=5\text{mm})=\dfrac{10^{-9}}{36\pi}(2)\dfrac{10^5}{5\ln3}=\underline{\underline{322\text{ nC/m}^2}}$

$\rho_s(\rho=15\text{mm})=-\dfrac{10^{-9}}{36\pi}(2)\dfrac{10^5}{15\ln3}=\underline{\underline{-107.3\text{ nC/m}^2}}$

Prob. 6.16

$\dfrac{1}{\rho}\dfrac{d^2V}{d\phi^2}=0\longrightarrow\dfrac{d^2V}{d\phi^2}=0\longrightarrow\dfrac{dV}{d\phi}=A$

$V=A\phi+B$

$0=0+B\longrightarrow B=0$

$50=A\pi/2\longrightarrow A=\dfrac{100}{\pi}$

$\boldsymbol{E}=-\nabla V=-\dfrac{1}{\rho}\dfrac{dV}{d\phi}\boldsymbol{a}_\phi=-\dfrac{A}{\rho}\boldsymbol{a}_\phi=\underline{\underline{-\dfrac{100}{\pi\rho}\boldsymbol{a}_\phi}}$

Prob. 6.29 If the centers at $\phi=0$ and $\phi=\pi/2$ are maintained at a potential difference of V_o, from Example 6.3,

$E_\phi=\dfrac{2V_o}{\pi\rho},\qquad J=\sigma E$

Hence,

$I=\int J\bullet dS=\dfrac{2V_o\sigma}{\pi}\int_{\rho=a}^{b}\int_{z=0}^{t}\dfrac{1}{\rho}d\rho dz=\dfrac{2V_o\sigma t}{\pi}\ln(b/a)$

and

$R=\dfrac{V_o}{I}=\dfrac{\pi}{2\sigma t\ln(b/a)}$

Prob. 6.30 If $V(r=a)=0$, $V(r=b)=V_o$, from Example 6.9,

$E=\dfrac{V_o}{r^2(1/a-1/b)},\qquad J=\sigma E$

Hence,

$I=\int J\bullet dS=\dfrac{V_o\sigma}{1/a-1/b}\int_{\theta=0}^{\alpha}\int_{\phi=0}^{2\pi}\dfrac{1}{r^2}r^2\sin\theta d\theta d\phi=\dfrac{2\pi V_o\sigma}{1/a-1/b}(-\cos\theta)|_0^\alpha$

$R=\dfrac{V_o}{I}=\dfrac{\frac{1}{a}-\frac{1}{b}}{2\pi\sigma(1-\cos\alpha)}$

Prob. 6.31

This is the same as Problem 6.30 except that $\alpha=\pi$. Hence,

$R=\dfrac{1}{2\pi\sigma(1-\cos\pi)}\left(\dfrac{1}{a}-\dfrac{1}{b}\right)=\underline{\dfrac{1}{4\pi\sigma}\left(\dfrac{1}{a}-\dfrac{1}{b}\right)}$

Prob. 6.66

4nC	-3nC		3nC	-4nC
-2	-1	0	1	2
4	3		2	1

(a) Q_i = -(3nC – 4nC) = <u>1nC</u>

(b) The force of attraction between the charges and the plates is

$\boldsymbol{F}=\boldsymbol{F}_{13}+\boldsymbol{F}_{14}+\boldsymbol{F}_{23}+\boldsymbol{F}_{24}$

$|F|=\dfrac{10^{-18}}{4\pi\times10^{-9}/36\pi}\left[\dfrac{9}{2^2}-\dfrac{2(12)}{3^2}+\dfrac{16}{4^2}\right]=\underline{\underline{5.25\text{ nN}}}$

6.1 Given $V = 5x^3y^2z$ and $\varepsilon = 2.25\varepsilon_o$, find (a) **E** at point $P(-3, 1, 2)$, (b) ρ_v at P.

6.2 Let $V = \dfrac{10\cos\theta\sin\phi}{r^2}$ and $\varepsilon = \varepsilon_o$. (a) Find **E** at point $P(1, 60°, 30°)$. (b) Determine ρ_v at P.

6.4 In free space, $V = 10\rho^{0.8}$ V. Find **E** and the volume charge density at $\rho = 0.6$ m.

6.7 In cylindrical coordinates, $V = 0$ at $\rho = 2$ m and $V = 60$ V at $\rho = 5$ m due to charge distribution $\rho_v = \dfrac{10}{\rho}$ pC/m³. If $\varepsilon_r = 3.6$, find **E**.

6.11 Given $V = x^3y + yz + cz^2$, find c such that V satisfies Laplace's equation.

Prob. 6.1

(a)

$$\boldsymbol{E} = -\nabla V = -\left(\frac{\partial V}{\partial x}\boldsymbol{a}_x + \frac{\partial V}{\partial z}\boldsymbol{a}_y + \frac{\partial V}{\partial z}\boldsymbol{a}_z\right)$$

$$= -(15x^2y^2z\boldsymbol{a}_x + 10x^3yz\boldsymbol{a}_y + 5x^3y^2\boldsymbol{a}_z)$$

At P, x=-3, y=1, z=2,

$$\boldsymbol{E} = -15(9)(1)(2)\boldsymbol{a}_x + 10(-27)(1)(2)\boldsymbol{a}_y - 5(-27)(1)\boldsymbol{a}_z = \underline{\underline{-270\boldsymbol{a}_x + 540\boldsymbol{a}_y + 135\boldsymbol{a}_z \text{ V/m}}}$$

(b) $\rho_v = \nabla \bullet \boldsymbol{D}$ or $\rho_v = -\varepsilon\nabla^2 V$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial y^2} = \frac{\partial}{\partial x}(15x^2y^2z) + \frac{\partial}{\partial y}(10x^3yz) + \frac{\partial}{\partial z}(5x^3y^2)$$

$$= 30xy^2z + 10x^3z$$

At P,

$$\rho_v = -\varepsilon\nabla^2 V = -2.25\times\frac{10^{-9}}{36\pi}\left[30(-3)(1)(2) + 10(-27)(2)\right] = \underline{14.324 \text{ nC/m}^3}$$

Prob. 6.2

(a)

$$-\boldsymbol{E} = \nabla V = -\frac{20}{r^3}\cos\theta\sin\phi\boldsymbol{a}_r - \frac{10}{r^3}\sin\theta\sin\phi\boldsymbol{a}_\theta + \frac{1}{r\sin\theta}\frac{10}{r^2}\cos\theta\cos\phi\boldsymbol{a}_\phi$$

At $P(1, 60°, 30°)$, $r = 1, \theta = 60°, \phi = 30°$

$$\boldsymbol{E} = \frac{20}{1^3}\cos 60°\sin 30°\boldsymbol{a}_r + \frac{10}{1^3}\sin 60°\sin 30°\boldsymbol{a}_\theta - \frac{1}{\sin 60°}\frac{10\cos 60°\cos 30°}{1^3}\boldsymbol{a}_\phi$$

$$= \underline{\underline{5\boldsymbol{a}_r + 4.33\boldsymbol{a}_\theta - 5\boldsymbol{a}_\phi \text{ V/m}}}$$

Prob. 6.4

$$\boldsymbol{E} = -\nabla V = -\frac{dV}{d\rho}\boldsymbol{a}_\rho = -(0.8)(10)\rho^{-0.2}\boldsymbol{a}_\rho = -8(0.6)^{-0.2}\boldsymbol{a}_\rho = \underline{\underline{-8.861\boldsymbol{a}_\rho}}$$

$$\nabla^2 V = \frac{1}{\rho}\frac{d}{d\rho}\left(\rho\frac{dV}{d\rho}\right) = -\frac{\rho_v}{\varepsilon_o} = \frac{1}{\rho}\frac{d}{d\rho}\left(8\rho^{0.8}\right) = \frac{1}{\rho}(8)0.8\rho^{-0.2} = 6.4\rho^{-1.2}$$

$$\rho_v = -\varepsilon_o\nabla^2 V = -6.4\times\frac{10^{-9}}{36\pi}\rho^{-1.2} = -\frac{6.4(0.6)^{-1.2}}{36\pi} \text{ nC/m}^3 = \underline{\underline{-0.1044 \text{ nC/m}^3}}$$

Prob. 6.7

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon} = \frac{-\dfrac{10}{\rho}\times 10^{-12}}{3.6\times\dfrac{10^{-9}}{36\pi}} = -\frac{0.1\pi}{\rho}$$

Let $\alpha = 0.1\pi$.

$$\nabla^2 V = -\frac{\alpha}{\rho} = \frac{1}{\rho}\frac{d}{d\rho}\left(\rho\frac{dV}{d\rho}\right)$$

$$-\alpha = \frac{d}{d\rho}\left(\rho\frac{dV}{d\rho}\right)$$

$$\rho\frac{dV}{d\rho} = -\alpha\rho + A$$

$$\frac{dV}{d\rho} = -\alpha + \frac{A}{\rho}$$

$$V = -\alpha\rho + A\ln\rho + B$$

At ρ=2, V=0 $\longrightarrow$ $0 = -2\alpha + A\ln 2 + B$ (1)

At ρ=5, V=60 $\longrightarrow$ $60 = -5\alpha + A\ln 5 + B$ (2)

Subtracting (1) from (2),

$$60 = -3\alpha + A\ln 5/2 \longrightarrow A = \frac{60 + 3\alpha}{\ln 2.5} = 66.51$$

From (1),

$$B = 2\alpha - A\ln 2 = -45.473$$

$$\boldsymbol{E} = -\frac{dV}{d\rho}\boldsymbol{a}_\rho = (\alpha - \frac{A}{\rho})\boldsymbol{a}_\rho = \underline{\underline{(0.3142 - \frac{66.51}{\rho})\boldsymbol{a}_\rho}}$$

Prob. 6.11

$$\nabla^2 U = \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = 6xy + 0 + 2c = 0$$

$$\underline{\underline{c = -3xy}}$$

5.38 Let $z < 0$ be region 1 with dielectric constant $\varepsilon_{r1} = 4$, while $z > 0$ is region 2 with $\varepsilon_{r2} = 7.5$. Given that $\mathbf{E}_1 = 60\mathbf{a}_x - 100\mathbf{a}_y + 40\mathbf{a}_z$ V/m, (a) find $\mathbf{P}_1$, (b) calculate $\mathbf{D}_2$.

5.39 Region 1 is $x < 0$ with, $\varepsilon_1 = 4\varepsilon_o$, while region 2 is $x > 0$ with $\varepsilon = 2\varepsilon_o$. If $\mathbf{E}_2 = 6\mathbf{a}_x - 10\mathbf{a}_y + 8\mathbf{a}_z$ V/m, (a) find $\mathbf{P}_1$, and $\mathbf{P}_2$, (b) calculate the energy densities in both regions.

5.40 A dielectric interface is defined by $4x + 3y = 10$ m. The region including the origin is free space, where $\mathbf{D}_1 = 2\mathbf{a}_x - 4\mathbf{a}_y + 6.5\mathbf{a}_z$ nC/m². In the other region, $\varepsilon_{r2} = 2.5$. Find $\mathbf{D}_2$ and the angle θ_2 that $\mathbf{D}_2$ makes with the normal.

5.41 Regions 1 and 2 have permittivities $\varepsilon_1 = 2\varepsilon_o$ and $\varepsilon_2 = 5\varepsilon_o$. The regions are separated by a plane whose equation is $x + 2y + z = 1$ such that $x + 2y + z > 1$ is region 1. If $\mathbf{E}_1 = 20\mathbf{a}_x - 10\mathbf{a}_y + 40\mathbf{a}_z$ V/m, find: (a) the normal and tangential components of $\mathbf{E}_1$, (b) $\mathbf{E}_2$.

5.42 Given that $\mathbf{E}_1 = 10\mathbf{a}_x - 6\mathbf{a}_y + 12\mathbf{a}_z$ V/m in Figure 5.22, find (a) $\mathbf{P}_1$, (b) $\mathbf{E}_2$ and the angle $\mathbf{E}_2$ makes with the y-axis, (c) the energy density in each region.

Prob. 5.38

(a)

$$\boldsymbol{P}_1 = \chi_{e1}\varepsilon_o\boldsymbol{E}_1 = 3\times\frac{10^{-9}}{36\pi}(60, -100, 40) = \underline{\underline{1.591\boldsymbol{a}_x - 2.6526\boldsymbol{a}_y + 1.061\boldsymbol{a}_z \text{ nC/m}^2}}$$

(b)

$$\boldsymbol{E}_{2t} = \boldsymbol{E}_{1t} = 60\boldsymbol{a}_x - 100\boldsymbol{a}_y$$

$$\boldsymbol{D}_{2n} = \boldsymbol{D}_{1n} \rightarrow \varepsilon_2\boldsymbol{E}_{2n} = \varepsilon_1\boldsymbol{E}_{1n}$$

$$\boldsymbol{E}_{2n} = \frac{\varepsilon_1}{\varepsilon_2}\boldsymbol{E}_{1n} = \frac{4}{7.5}(40\boldsymbol{a}_z) = 21.33\boldsymbol{a}_z$$

$$\boldsymbol{E}_2 = 60\boldsymbol{a}_x - 100\boldsymbol{a}_y + 21.33\boldsymbol{a}_z$$

$$\boldsymbol{D}_2 = \varepsilon_o\varepsilon_{r2}\boldsymbol{E}_2 = 7.5\times\frac{10^{-9}}{36\pi}(60, -100, 21.33) = \underline{\underline{3.979\boldsymbol{a}_x - 6.631\boldsymbol{a}_y + 1.414\boldsymbol{a}_z \text{ nC/m}^2}}$$

Prob. 5.39

(a)

$$\boldsymbol{E}_{2t} = \boldsymbol{E}_{1t} = -10\boldsymbol{a}_y + 8\boldsymbol{a}_z$$

$$\boldsymbol{D}_{2n} = \boldsymbol{D}_{1n} \rightarrow \varepsilon_2\boldsymbol{E}_{2n} = \varepsilon_1\boldsymbol{E}_{1n}$$

$$\boldsymbol{E}_{1n} = \frac{\varepsilon_2}{\varepsilon_1}\boldsymbol{E}_{2n} = \frac{2\varepsilon_o}{4\varepsilon_o}(6\boldsymbol{a}_x) = 3\boldsymbol{a}_x$$

$$\boldsymbol{E}_1 = 3\boldsymbol{a}_x - 10\boldsymbol{a}_y + 8\boldsymbol{a}_z$$

$$\boldsymbol{P}_1 = \chi_{e1}\varepsilon_o\boldsymbol{E}_1 = 3\times\frac{10^{-9}}{36\pi}(3, -10, 8) = \underline{\underline{79.6\boldsymbol{a}_x - 265.3\boldsymbol{a}_y + 212.2\boldsymbol{a}_z \text{ nC/m}^2}}$$

$$\boldsymbol{P}_2 = \chi_{e2}\varepsilon_o\boldsymbol{E}_2 = 1\times\frac{10^{-9}}{36\pi}(6, -10, 8) = \underline{\underline{53.05\boldsymbol{a}_x - 88.42\boldsymbol{a}_y + 70.74\boldsymbol{a}_z \text{ nC/m}^2}}$$

(b)

$$w_1 = \frac{1}{2}\varepsilon_1 E_1^2 = \frac{1}{2}(4\varepsilon_o)(9 + 100 + 64) = 346\varepsilon_o = \underline{\underline{3.0593 \text{ nJ/m}^2}}$$

$$w_2 = \frac{1}{2}\varepsilon_2 E_2^2 = \frac{1}{2}(2\varepsilon_o)(36 + 100 + 64) = 200\varepsilon_o = \underline{\underline{1.7684 \text{ nJ/m}^2}}$$

Prob. 5.40

f(x,y)= 4x +3y −10=0

$$\nabla f = 4\boldsymbol{a}_x + 3\boldsymbol{a}_y \longrightarrow \boldsymbol{a}_n = -\frac{\nabla f}{|\nabla f|} = \frac{-(4\boldsymbol{a}_x + 3\boldsymbol{a}_y)}{5} = -0.8\boldsymbol{a}_x - 0.6\boldsymbol{a}_y$$

The minus sign is chosen for $\mathbf{a}_n$ because it is directed toward the origin.

$$\boldsymbol{D}_{1n} = (\boldsymbol{D}_1\cdot\boldsymbol{a}_n)\boldsymbol{a}_n = (1.6 - 2.4)\boldsymbol{a}_n = -0.64\boldsymbol{a}_x - 0.48\boldsymbol{a}_y$$

$$\boldsymbol{D}_{1t} = \boldsymbol{D}_1 - \boldsymbol{D}_{1n} = 2.64\boldsymbol{a}_x - 3.52\boldsymbol{a}_y + 6.5\boldsymbol{a}_z$$

$$\boldsymbol{D}_{2n} = \boldsymbol{D}_{1n} = -0.64\boldsymbol{a}_x - 0.48\boldsymbol{a}_y$$

$$\boldsymbol{E}_{2t} = \boldsymbol{E}_{1t} \longrightarrow \frac{\boldsymbol{D}_{2t}}{\varepsilon_2} = \frac{\boldsymbol{D}_{1t}}{\varepsilon_2}$$

Prob. 5.41

(a)

Let $f(x, y, z) = x + 2y + z - 1 = 0$

$$\nabla f = \boldsymbol{a}_x + 2\boldsymbol{a}_y + \boldsymbol{a}_z \rightarrow \boldsymbol{a}_n = \frac{\nabla f}{|\nabla f|} = \frac{1}{\sqrt{6}}(\boldsymbol{a}_x + 2\boldsymbol{a}_y + \boldsymbol{a}_z)$$

$$\boldsymbol{E}_{1n} = (\boldsymbol{E}_1\bullet\boldsymbol{a}_n)\boldsymbol{a}_n = \frac{1}{\sqrt{6}}(20 - 20 + 40)\frac{1}{\sqrt{6}}(\boldsymbol{a}_x + 2\boldsymbol{a}_y + \boldsymbol{a}_z) = \underline{\underline{6.667\boldsymbol{a}_x + 13.33\boldsymbol{a}_y + 6.667\boldsymbol{a}_z \text{ V/m}}}$$

$$\boldsymbol{E}_{1t} = \boldsymbol{E}_1 - \boldsymbol{E}_{1n} = \underline{\underline{13.3\boldsymbol{a}_x - 23.3\boldsymbol{a}_y + 33.3\boldsymbol{a}_z \text{ V/m}}}$$

(b)

$$\boldsymbol{E}_{2t} = \boldsymbol{E}_{1t} = 13.3\boldsymbol{a}_x - 23.3\boldsymbol{a}_y + 33.3\boldsymbol{a}_z$$

$$\boldsymbol{D}_{2n} = \boldsymbol{D}_{1n} \rightarrow \varepsilon_2\boldsymbol{E}_{2n} = \varepsilon_1\boldsymbol{E}_{1n}$$

$$\boldsymbol{E}_{2n} = \frac{\varepsilon_1}{\varepsilon_2}\boldsymbol{E}_{1n} = \frac{2}{5}(6.667, 13.33, 6.667) = 2.7\boldsymbol{a}_x + 5.3\boldsymbol{a}_y + 33.3\boldsymbol{a}_z$$

$$\boldsymbol{E}_2 = \boldsymbol{E}_{2t} + \boldsymbol{E}_{2n} = \underline{\underline{16\boldsymbol{a}_x - 18\boldsymbol{a}_y + 36\boldsymbol{a}_z \text{ V/m}}}$$

Prob. 5.42

(a) $\boldsymbol{P}_1 = \varepsilon_o\chi_{e1}\boldsymbol{E}_1 = 2\times\dfrac{10^{-9}}{36\pi}(10, -6, 12) = \underline{\underline{0.1768\boldsymbol{a}_x - 0.1061\boldsymbol{a}_y + 0.2122\boldsymbol{a}_z \text{ nC/m}^2}}$

(b) $\boldsymbol{E}_{1n} = -6\boldsymbol{a}_y$, $\quad \boldsymbol{E}_{2t} = \boldsymbol{E}_{1t} = 10\boldsymbol{a}_x + 12\boldsymbol{a}_z$

$$\boldsymbol{D}_{2n} = \boldsymbol{D}_{1n} \longrightarrow \varepsilon_2\boldsymbol{E}_{2n} = \varepsilon_1\boldsymbol{E}_{1n}$$

or $\boldsymbol{E}_{2n} = \dfrac{\varepsilon_1}{\varepsilon_2}\boldsymbol{E}_{1n} = \dfrac{3\varepsilon_o}{4.5\varepsilon_o}(-6\boldsymbol{a}_z) = -4\boldsymbol{a}_y$

$$\boldsymbol{E}_2 = \underline{\underline{10\boldsymbol{a}_x - 4\boldsymbol{a}_y + 12\boldsymbol{a}_z \text{ V/m}}}$$

$$\tan\theta_2 = \frac{E_{2t}}{E_{2n}} = \frac{\sqrt{10^2 + 12^2}}{4} = 3.905 \longrightarrow \underline{\underline{\theta_2 = 75.64°}}$$

(c) $w_E = \dfrac{1}{2}\boldsymbol{D}\bullet\boldsymbol{E} = \dfrac{1}{2}\varepsilon|\boldsymbol{E}|^2$

$$w_{E1} = \frac{1}{2}\varepsilon_1|\boldsymbol{E}_1|^2 = \frac{1}{2}\text{x}3\text{x}\frac{10^{-9}}{36\pi}(10^2 + 6^2 + 12^2) = \underline{\underline{3.7136 \text{ nJ/m}^3}}$$

$$w_{E2} = \frac{1}{2}\varepsilon_2|\boldsymbol{E}_2|^2 = \frac{1}{2}\text{x}4.5\text{x}\frac{10^{-9}}{36\pi}(10^2 + 4^2 + 12^2) = \underline{\underline{5.1725 \text{ nJ/m}^3}}$$